Headline XVII: the random per-stratum posterior expectation in general dimension

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2026-09-07

Abstract

Unit 213 combines Headline XVI with the positive-denominator quotient theorem of unit 157. With a deterministic continuous observable φ and density $c > 0$ on the closed cube, a random phase $\xi_n : \Omega \to C([0,1]^d)$ converging in distribution to ξ, and deterministic scales $N_n \to \infty$ ($N_n > 1$),

$$\frac{\int \varphi c\, u^h e^{-\beta N_n^2 u^{2k} + \beta N_n u^k \xi_n}\, du}{\int c\, u^h e^{-\beta N_n^2 u^{2k} + \beta N_n u^k \xi_n}\, du} \Rightarrow \frac{\mathrm{phaseCoeff}(\xi, \varphi c)}{\mathrm{phaseCoeff}(\xi, c)}$$

(`tendstoInDistribution_phase_posterior`); in the equal-ratio case the limit is the deterministic corner value $\varphi(0)$, whatever the limiting phase (`tendstoInDistribution_phase_posterior_equal`) — the general-d form of the $d = 2$ chart posterior limit $Z[\varphi]/Z[1] \Rightarrow y_{\varphi,00}/y_{1,00}$. Review v12 (units 209–212): clean at statement level, prose should-fixes applied. Zero `sorry`/`axiom`; 9090 jobs.

1 The things formalised

```
instance : MeasurableSpace C(closedCube d, ) := borel _   (BorelSpace; product =
      Borel on InputSpace)
theorem measurable_pair_const (hX : Measurable X) (g) : Measurable fun  => ((X  ,
      g) : InputSpace d)
theorem continuous_pair_const (g) : Continuous fun f => ((f, g) : InputSpace d)
theorem extCube_mul (  c) : extCube (  * c) = fun x => extCube   x * extCube c x
theorem tendstoInDistribution_normChart_pair … (g  g ) :
    TendstoInDistribution (fun n   => (normChart … (Nseq n) (X n  , g ), normChart
      … (Nseq n) (X n  , g ))) L
      (fun   => (limChart … (Z  , g ), limChart … (Z  , g ))) (fun _ =>  )  '
theorem tendstoInDistribution_phase_posterior … (hN1 :   n, 1 < Nseq n) (  c)
      (hcpos :   x, 0 < c x) :
    TendstoInDistribution (fun n   => chartIntegral … (Nseq n) (extCube (X n  ))
      (extCube (  * c))
        / chartIntegral … (Nseq n) (extCube (X n  )) (extCube c)) L
      (fun   => limChart … (Z  ,   * c) / limChart … (Z  , c)) (fun _ =>  )  '
theorem tendstoInDistribution_phase_posterior_equal (hratio) … : … L (fun _ =>   0)
      (fun _ =>  )  '
```

2 Mathematics

The numerator and denominator are functions of the same random phase paired with the fixed amplitudes φc and c, so their joint convergence is Headline XVI for the vector $(F_N(\cdot, \varphi c), F_N(\cdot, c))$:

the pair of errors tends to 0 in probability (`tendstoInMeasure_prodMk_zero`) and the pair of limits is a continuous function of the phase. The limiting denominator phaseCoeff(ξ, c) is strictly positive for every ξ by unit 205 (it depends on ξ only through $J_p(\xi(\pi u)) > 0$), so the quotient theorem applies with totalised division; the common scale cancels exactly once $N_n > 1$, which is why that hypothesis replaces $N_n \geq 0$.

3 Lean notes

- Putting the Borel -algebra on the *factor* $C([0,1]^d, \mathbb{R})$ (rather than on the product) makes `Measurable.prodMk` available for pairing with a constant and gives `BorelSpace (InputSpace d)` through `Prod.borelSpace`; a separate `borel` instance on the product would create an instance diamond.
- `TendstoInDistribution.congr` needs a.e. equality for *every* index, so the totalised quotient identity $(a/S)/(b/S) = a/b$ must hold at every n; requiring $N_n > 1$ (rather than eventually) keeps $S \neq 0$ throughout.
- The corner point $\langle 0, \_ \rangle$ of the closed cube is not syntactically the clamp of 0; `Subtype.ext (clampCube_of_mem …)` bridges them.

4 Review v12 disposition

Units 209–212 clean. Applied: the unit-212 module doc now displays the ambient assumptions (probability measures, countably generated filter, deterministic $N_n \geq 0$), states that scale-dependent amplitudes are allowed when encoded in the input, and fixes the dimension mismatch; the mirror comparison is narrowed to a conditional leading-order analogue of the empirical-phase transfer.

5 Status

Headlines XIII–XVII: deterministic and random leading-order theory of one normal block in general dimension, plus conditional chart assembly. Next: Astra #23 for the remaining direction (signed orthants with phase, Mellin dictionary, or hand-off with report v3).